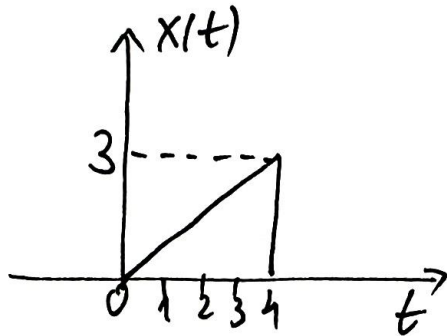


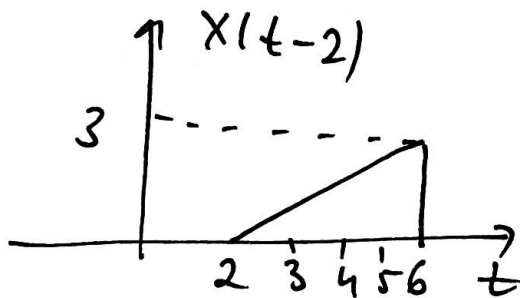
1.1 Për një sinjalin kontinual $x(t)$ të paraqitur në figurë
 shkoni sinjalet në vazhdim:

- a) $x(t-2)$ b) $x(2t)$ c) $x(t/2)$ d) $x(-t)$

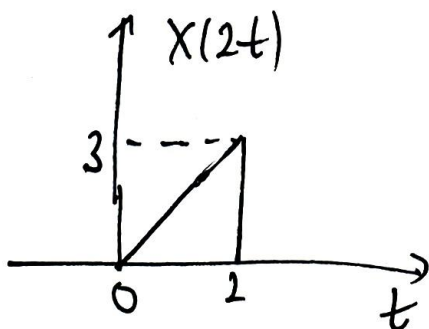


Zgjidhje

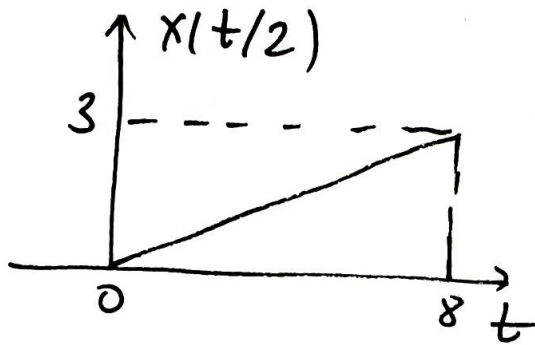
- a) Zhvendosje për 2 njësi djathtas:



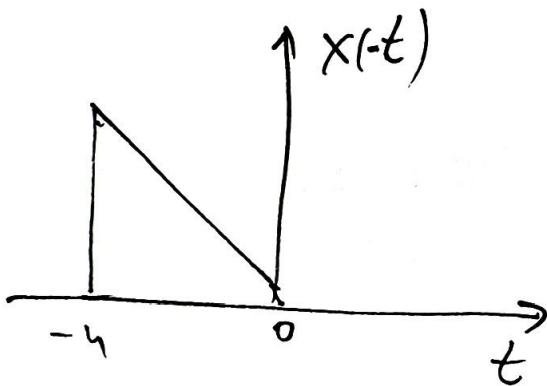
- b) Ngusëtim të brezit të sinjalit për 2 njësi



c) Zgjerim për 2 njës.

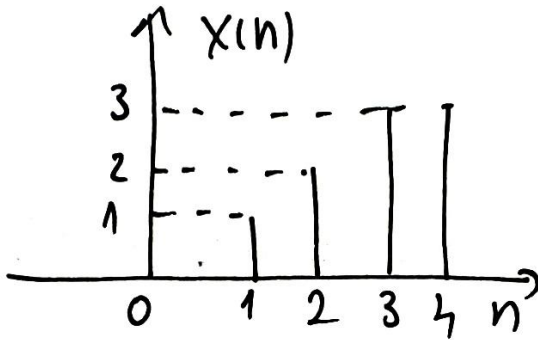


d) Pasqymim

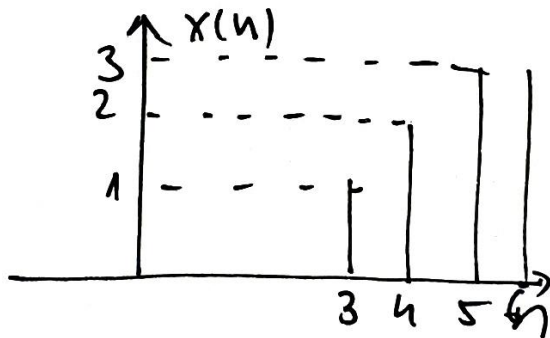


1.2 Për shqiptimin diskret të dhënë në figurë
shkruani shprehjet në vazhdim:

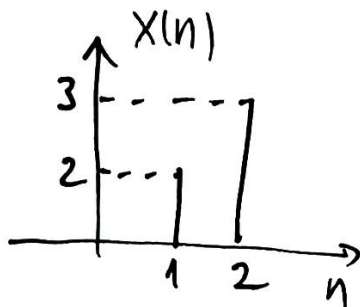
- a) $x(n-2)$ b) $x(2n)$ c) $x(-n)$ d) $x(-n+2)$



- a) Shprehjet për 2 njësi të shprehjes:

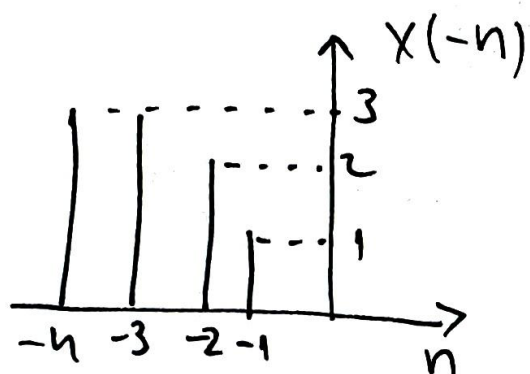


- b) Shprehjet për 2 njësi

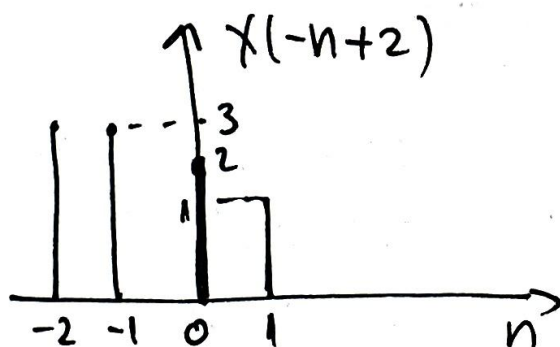
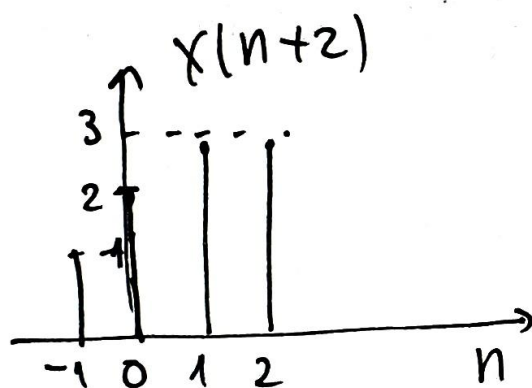


$\Rightarrow$ Shprehjet vetëm
shprehjet që dalin
me vlera të plota!

c) Pasgrynim



d) Kemi dy veti — žhrendosje majtas pēn 2
Pasgrynim



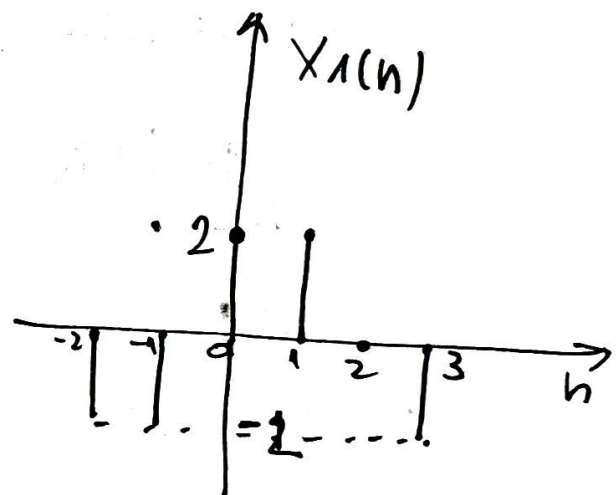
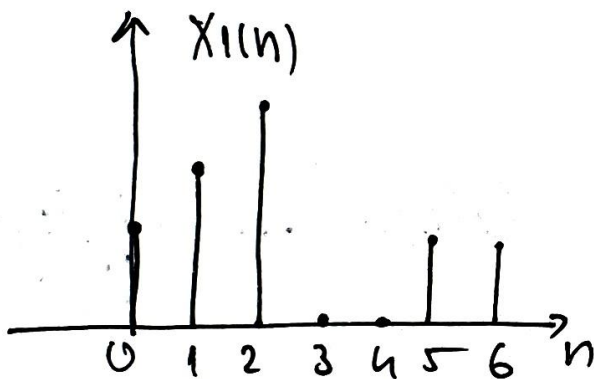
1.4 Duke përdorur singulet e dhëna diskrte $x_1[n]$ dhe $x_2[n]$ të paraqitura në fig.

Gjeni:

a) $y_1[n] = x_1[n] + x_2[n]$

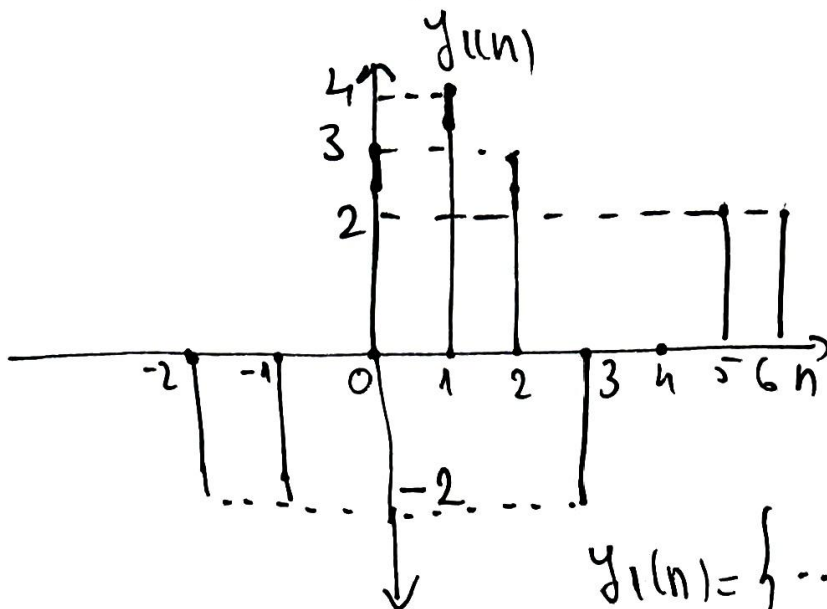
b) $y_2[n] = 2x_1[n]$

c) $y_3[n] = x_1[n] \cdot x_2[n]$



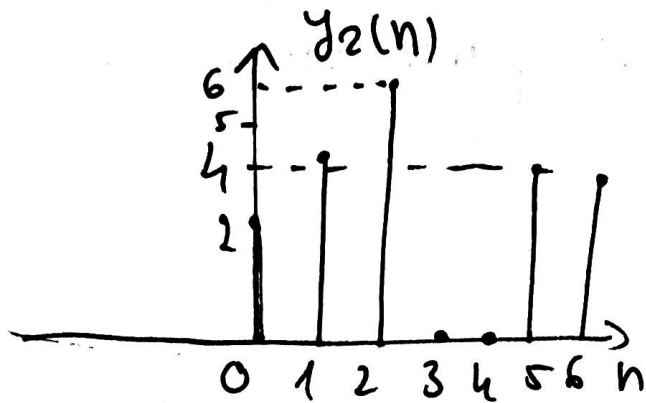
Zgjidhje

a) Mbledhim amplitudet marrin për marrjen



$$y_1[n] = \{ \dots, 0, -2, -2, 3, 4, 3, -2, 0, 2, 2.0 \}$$

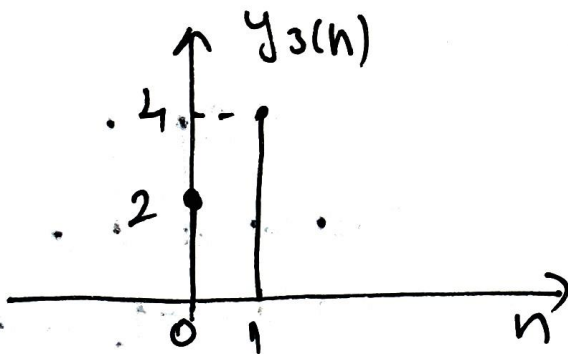
b) Shumzojmi amplituden e signalit $x_1(n)$ me 2.



$$y_2(n) = 2x_1(n)$$

$$y_1(n) = \{2, 4, 6, 0, 0, 4, 4\}$$

c) $y_3(n) = x_1(n) x_2(n)$



$$y_3(n) = \{\dots, 0, 2, 4, 0\}$$

↑

Singulet çiftë dhe tekë

Singuli çift

$$X(-t) = X(t)$$

$$X(-n) = X(n)$$

$$X_e(t) = \frac{X(t) + X(-t)}{2}$$

$$X_e(n) = \frac{X(n) + X(-n)}{2}$$

Singuli tekë

$$X(-t) = -X(t)$$

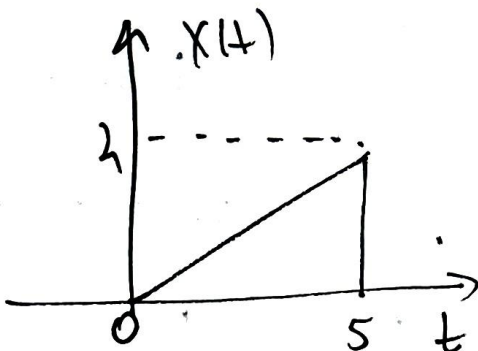
$$X(-n) = -X(n)$$

$$X_o(t) = \frac{X(t) - X(-t)}{2}$$

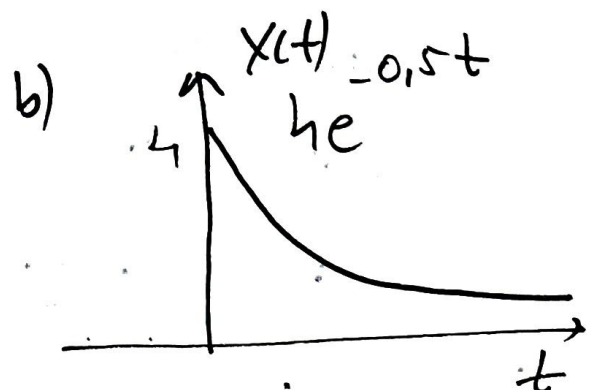
$$X_o(n) = \frac{X(n) - X(-n)}{2}$$

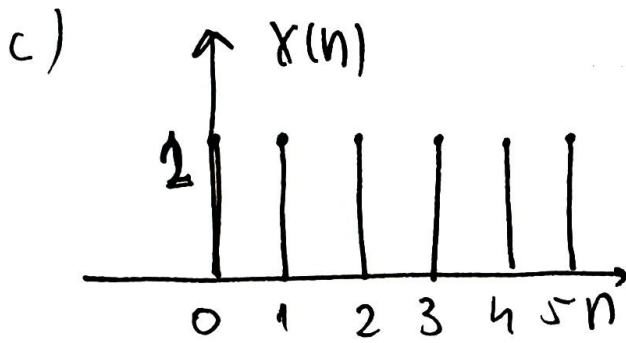
1.5 Shkoni komponentet çiftë dhe tekë të rrethave në figurë:

a)

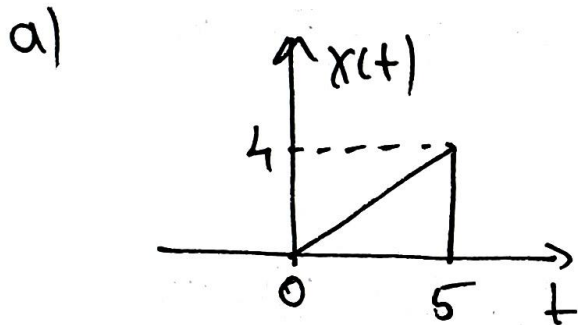
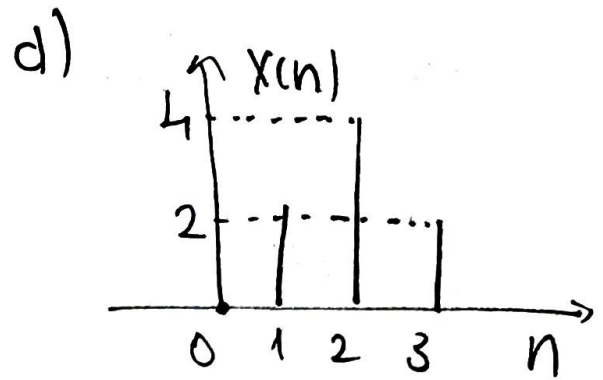


b)

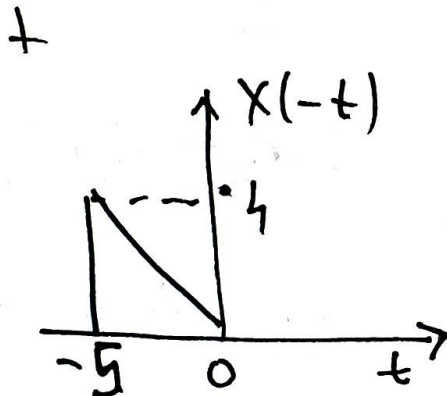




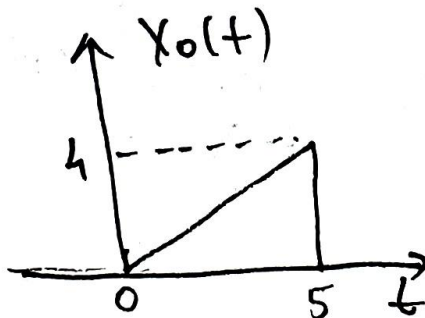
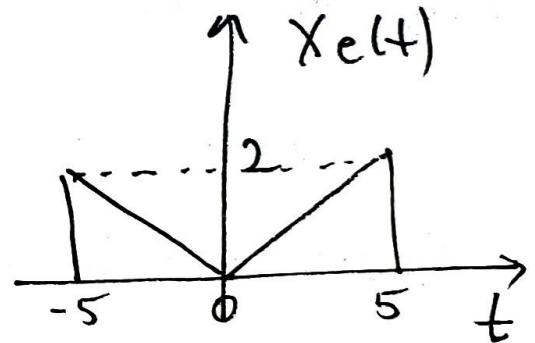
Ëqivalenc



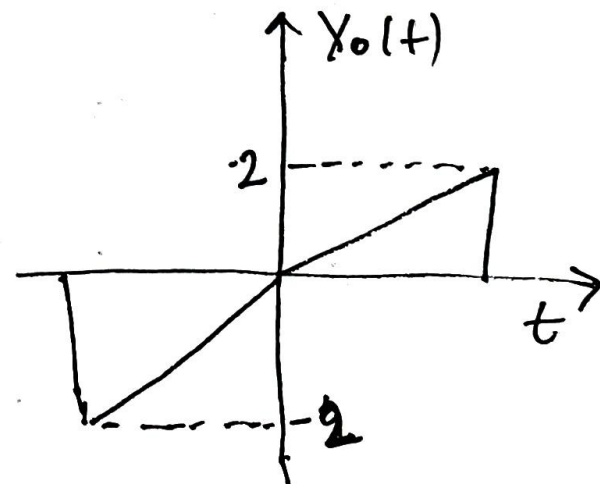
Pjesa e Gifte



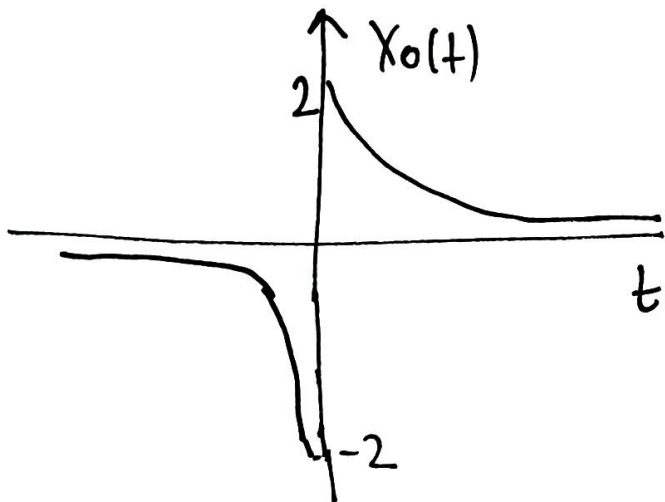
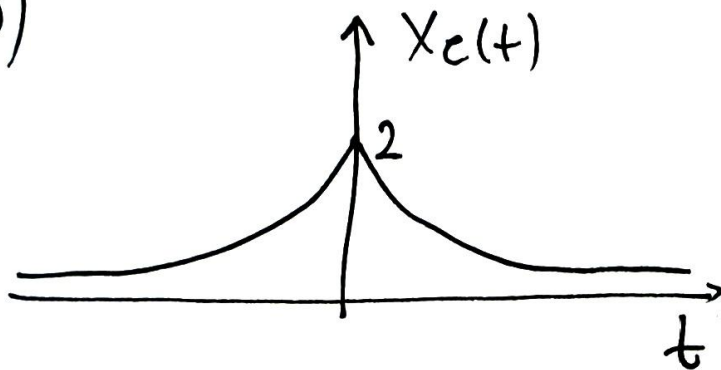
$\Rightarrow$



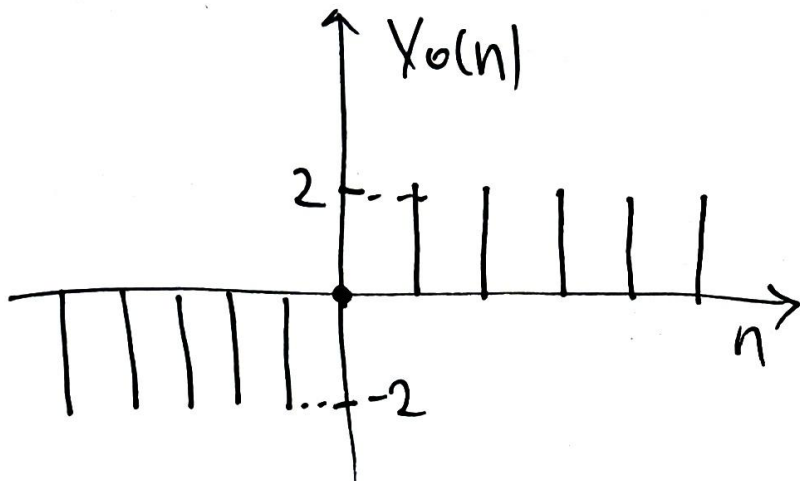
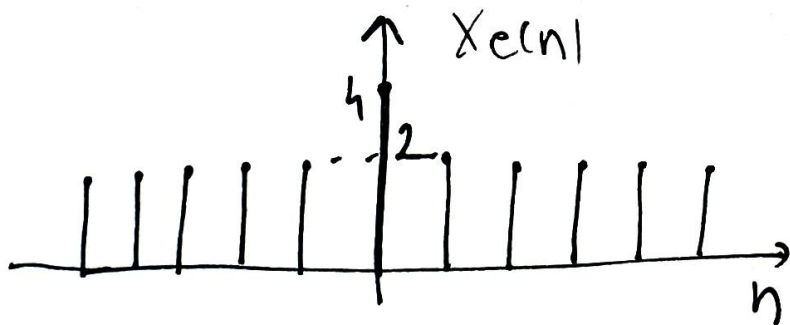
$\Rightarrow$

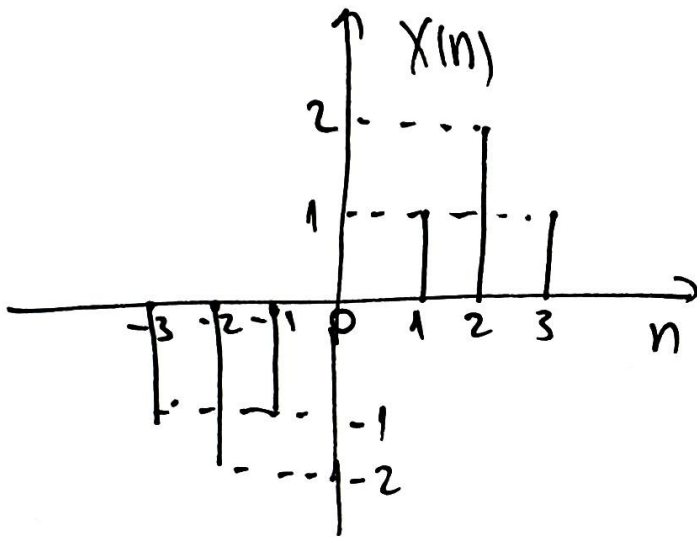
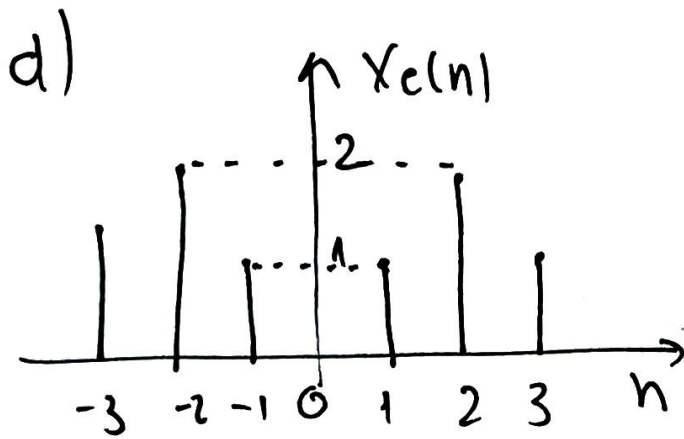


b)



c)





1.6 Gjeni pjesët qitë dhe të reë ringjalt
 $x(t) = e^{jt}$

$$e^{jt} = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{e^{jt} + e^{-jt}}{2} = \cos t$$

$$x_o(t) = \frac{e^{jt} - e^{-jt}}{2} = j \sin t$$

Formulat
 e
 Eulerit

Periodiciteti i singuleve

Singulet $\begin{cases} \text{periodike} \\ \text{jopriodike} \end{cases}$

* Singjali i vazhduar $x(t)$ është periodik nëse mund të gjendet në paku një $T \in \mathbb{R}$ për të cilin vlen:

$$x(t) = x(t+T)$$

* Singjali diskret $x(n)$ është periodik nëse mund të gjendet në paku një numër i plotë $N \in \mathbb{Z}$ për të cilin vlen:

$$x(n) = x[n+N]$$

1.10 $x(t) = \cos(\omega_0 t + \theta)$

Vërteto që perioda e këtij singjali është

$$\frac{2\pi}{\omega_0} :$$

$$\begin{aligned} \cos(\omega_0(t+T) + \theta) &= \cos(\omega_0 t + \theta + \omega_0 T) \\ &= \cos(\omega_0 t + \theta) \end{aligned}$$

Nëse $\omega_0 T = m 2\pi \Rightarrow T = m \frac{2\pi}{\omega_0} \rightarrow \text{Perioda}$

1.11. $X(n) = e^{j\Omega_0 n}$

Võrdeto ge' mingali e'htke' periodike' v'etm
nõm $\Omega_0/2\pi$ e'htke' numõr r'acional:

$$e^{j\Omega_0(n+N)} = e^{j\Omega_0 n} e^{j\Omega_0 N} = e^{j\Omega_0 n}$$

$$\Omega_0 N = m 2\pi$$

$$\frac{\Omega_0}{2\pi} = \frac{m}{N}$$

1.16

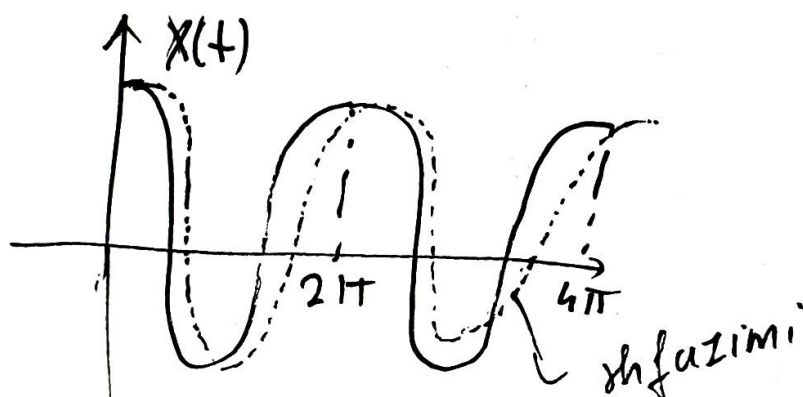
a) $X(t) = \cos\left(t + \frac{\pi}{4}\right)$

$\omega_0 = 1$

$\varphi = \frac{\pi}{4}$

$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{1} = 2\pi$

↓
signali
periodik

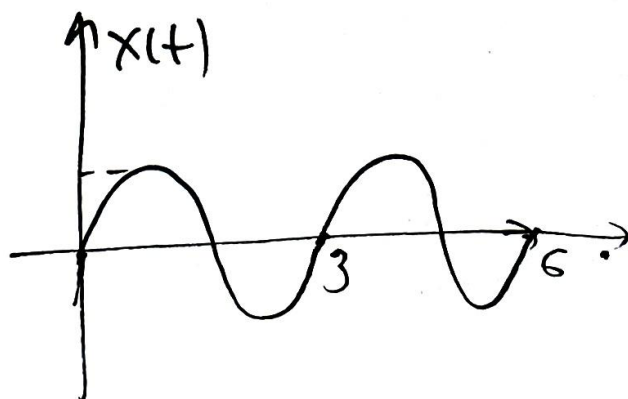


b)

$X(t) = \sin \frac{2\pi}{3} t$

$\omega_0 = \frac{2\pi}{3} \Rightarrow T = \frac{2\pi}{\omega_0} = \frac{2\pi}{\frac{2\pi}{3}} = \frac{2\pi \cdot 3}{2\pi} = 3$

↓
signali
periodik



$$c) \quad x(t) = \cos \frac{\pi}{3} t + \sin \frac{\pi}{4} t = x_1(t) + x_2(t)$$

$$\omega_1 = \frac{\pi}{3} \Rightarrow T_1 = \frac{2\pi}{\frac{\pi}{3}} = 6$$

$$\omega_2 = \frac{\pi}{4} \Rightarrow T_2 = \frac{2\pi}{\frac{\pi}{4}} = 8$$

$$\frac{T_1}{T_2} = \frac{6:2}{8:2} = \frac{3}{4}$$

$$T_1 \cdot 4 = T_2 \cdot 3$$

$$6 \cdot 4 = 8 \cdot 3$$

$$\boxed{24 = 24}$$

→ perioda themelone
singali periodik

$$d) \quad x(t) = \cos t + \sin \sqrt{2} t = x_1(t) + x_2(t)$$

$$\omega_1 = 1 \Rightarrow T = \frac{2\pi}{\omega_1} = 2\pi$$

$$\omega_2 = \sqrt{2} \Rightarrow T_2 = \frac{2\pi}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{2\pi\sqrt{2}}{2} = \sqrt{2}\pi$$

$$\frac{T_1}{T_2} = \frac{2\pi}{\sqrt{2}\pi} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \sqrt{2} \rightarrow \text{singali } \text{periodik}$$

Diskrete

a) $x(n) = \cos \frac{1}{4} n$

$$\Omega_0 = \frac{1}{4}$$

$$\frac{\Omega_0}{2\pi} = \frac{\frac{1}{4}}{2\pi} = \frac{1}{8\pi} \rightarrow \text{singulär periodisch}$$

b) $x(n) = \cos \frac{\pi}{3} n + \sin \frac{\pi}{4} n = x_1(n) + x_2(n)$

$$x_1(n) = \cos \frac{\pi}{3} n \Rightarrow \Omega_1 = \frac{\pi}{3} \Rightarrow \frac{\frac{\pi}{3}}{2\pi} = \frac{1}{6}$$

$$\boxed{N_1 = 6}$$

$$x_2(n) = \sin \frac{\pi}{4} n \Rightarrow \Omega_2 = \frac{\pi}{4} \Rightarrow \frac{\frac{\pi}{4}}{2\pi} = \frac{1}{8}$$

$$\boxed{N_2 = 8}$$

$$\frac{N_1}{N_2} = \frac{6:2}{8:2} = \frac{3}{4}$$

$$N_1 \cdot 4 = N_2 \cdot 3$$

$$6 \cdot 4 = 8 \cdot 3$$

$$\boxed{24 = 24}$$

singulär periodisch ←

$$c) \quad X(n) = \cos^2 \frac{\pi}{8} n$$

$$\boxed{\cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)}$$

$$X(n) = \frac{1}{2} (1 + \cos 2 \frac{\pi}{8} n)$$

$$= \frac{1}{2} (1 + \cos \frac{\pi}{4} n) = \underbrace{\frac{1}{2}}_{X_1} + \underbrace{\frac{1}{2} \cos \frac{\pi}{4} n}_{X_2}$$

$$X_1(n) = \frac{1}{2} \Rightarrow \frac{1}{2} \cdot (1)^n = \frac{1}{2} \quad \boxed{N_1 = 1}$$

$$X_2(n) = \frac{1}{2} \cos \frac{\pi}{4} n = \frac{1}{2} \cos \Omega_0 n$$

$$\frac{\Omega_0}{2\pi} = \frac{\frac{\pi}{4}}{2\pi} = \frac{\cancel{1}\pi}{8\cancel{\pi}} = \frac{1}{8} \rightarrow$$

macronal

$$\boxed{N_2 = 8}$$